

The risks of copper toxicity contributing to cognitive decline in the aging population and to Alzheimer's disease.

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It is a pleasure and an honor to contribute a paper to a special issue of the Journal of the American College of Nutrition honoring Stanley Wallach and Pearl Small. In this brief review I advance the hypothesis that copper toxicity is the major cause of the epidemic of mild cognitive impairment and Alzheimer's disease engulfing our aging population. This epidemic is recent, exploding in the last 50-60 years. The disease was virtually unknown 100 years ago. And it involves only developed countries that use copper plumbing. Something in our environment associated with development is poisoning the minds of our aged. The epidemic is associated with the use of copper plumbing, and the taking of copper in multi-mineral supplements. Food copper (organic copper) is processed by the liver and is transported and sequestered in a safe manner. Inorganic copper, such as that in drinking water and copper supplements, largely bypasses the liver and enters the free copper pool of the blood directly. This copper is potentially toxic because it may penetrate the blood/brain barrier. I review a web of animal and human data that tightens the noose around the hypothesis that copper toxicity is causing the epidemic of Alzheimer's disease and loss of cognition in our aging population.